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Generation of Thai Spelling Distributions for Reading through the improvement of Natural Language Processing Techniques

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***Abstract*— Thai spelling distributions for reading are fundamental to understanding word components and the principles of phoneme synthesis, which are essential for accurate pronunciation and reading comprehension. This research focuses on generating Thai spelling distributions, addressing the unique challenges posed by the complexity of Thai words. These challenges include the varying number of syllables, the intricate relationships among syllable components, and the non-linear arrangement of consonants and vowels. This research proposes a methodology for generating Thai spelling distributions for reading using improved natural language processing (NLP) techniques. The approach integrates a customized dictionary of Thai consonant and vowel sequences with statistical calculations based on the Normalized Levenshtein Distance (NLD). The system evaluates a dataset of authentic Thai words sourced from the Grade 1 Basic Vocabulary List provided by the Office of the Basic Education Commission. The proposed methodology demonstrated high accuracy, closely matching human-generated spelling distribution. This research also contributes a method for generating Thai spelling distribution and a novel framework for decomposing Thai word components. This framework holds significant potential for broader applications in Thai NLP tasks. It can support children with spelling difficulties, such as those with dyslexia, by creating educational tools and customized learning programs. These resources can improve spelling and reading skills for children with special needs, making language learning more inclusive and effective.**

Keywords—Thai spelling distributions for reading, formatting, Natural Language Processing (NLP), Customized dictionary, Normalized Levensthein Distance (NLD), Thai word decomposition

I. INTRODUCTION

Reading through spelling distribution is fundamental for breaking down syllables into their constituent sounds, including initial consonants, vowels, tonal markers, and final consonants. This process is crucial for learning the principles of Thai pronunciation and is widely used in early education. Spelling distribution, in particular, refers to the systematic decomposition of Thai words into phonetic components, allowing learners to recognize patterns in pronunciation. Due to its significance, spelling distribution has been incorporated into the Thai Basic Education Core Curriculum B.E. 2551 (2008) for first-grade students to help them read fundamental words commonly used in daily life.

According to an interview with an expert in special education and inclusive learning from the Faculty of Education, Chulalongkorn University, phonetic segmentation is actively taught in first-grade classrooms, with individual student assessments. However, due to limitations in the teacher-to-student ratio, comprehensive student engagement remains challenging. Studies have shown that large class sizes can hinder effective phonetic instruction, leading to varied student learning outcomes. Furthermore, individuals outside formal education, such as foreign learners attempting to study Thai through phonetic segmentation, face difficulties due to a lack of instructional materials tailored to this approach. Therefore, it is essential to generate a dataset of Thai words paired with their segmented phonetic breakdowns. Such a dataset would facilitate self-learning for interested individuals and serve as a foundation for developing tools to assess phonetic segmentation skills in the future.

Thai exhibits a complex orthographic structure, posing challenges in phonetic segmentation. A single Thai word may consist of one or multiple syllables, depending on its phonemic composition. For example, the word ตะเกียบ (chopsticks) can be segmented into ตะ | เกียบ. The arrangement of consonants and vowels within syllables varies significantly, along with tonal markers and their corresponding tonal modifications. Furthermore, depending on its syllabic position, a single character may function as an initial consonant, vowel, or final consonant. For instance, in the word ออม (economize), the first occurrence of อ serves as an initial consonant, while the second occurrence functions as a vowel.

Additionally, character clusters often comprise multiple characters, further complicating phonetic segmentation. These complexities make generating Thai word pairs with their corresponding phonetic segmentation challenging. This challenge includes syllable segmentation and character classification within a syllable into initial consonants, vowels, final consonants, and tonal markers, followed by phonetic mapping to construct distributed spelling transcriptions.

PyThaiNLP [1] has developed tokenization algorithms for Thai text segmentation, including sentence segmentation, word segmentation, and syllable tokenization [2]. The syllable tokenization method presented in [2] employs a trigram-based model trained on a corpus of over 3.1 million Thai words, which can be leveraged for this research. However, while existing models achieve high word and syllable segmentation accuracy, they do not explicitly generate structured spelling distributions aligned with phonetic segmentation principles. This research aims to bridge this gap by developing a

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specialized method that improves Thai spelling distribution generation.

Levensthein distance, a widely used NLP tool, has been employed to measure the edit distance between word pairs, reflecting their phonetic similarity. It has also been applied in orthographic analysis to distinguish words across dialectal variations [3]. Additionally, in [4], the Levensthein algorithm was used to refine Arabic transliterations from Arabizi by assigning different weights to commonly substituted characters (e.g., "3" → "ع") and retrieving the closest matching words from an Arabic dictionary, thereby enhancing transliteration accuracy. In this research, we adapt the Normalized Levensthein Distance (NLD) to enhance Thai spelling distribution generation by calculating phonetic proximity within a predefined Thai consonant and vowel sequence dictionary. This approach improves segmentation accuracy and provides a structured representation of Thai spelling distributions.

The results of this research have significant implications for Thai NLP applications, particularly in pronunciation assessment, Thai language learning tools, and speech synthesis. By increasing the precision and effectiveness of Thai word component decomposition, this study contributes to the generation of Thai spelling distributions and broader advancements in Thai language processing and educational technology.

II. Related work

A. Composition of Thai Words

Thai words are known for their complex orthographic structure, where spaces do not explicitly delimit words and can consist of one or multiple syllables or phonemic units. Each syllable is formed by blending initial consonants, vowels, final consonants, and tonal markers, creating a non-linear composition in which the positional role of each character may vary depending on its surrounding context. For instance, in the word เกาะ (island), the character ะ functions as a vowel. In contrast, in กะเพรา (holy basil), the same character serves a different phonetic role within the syllabic structure. Similarly, Thai diphthongs and vowel clusters further contribute to the complexity, as their pronunciation and representation can shift based on adjacent consonants. This intricate interplay of phonetic elements necessitates advanced computational models capable of accurately segmenting and analyzing Thai words for various natural language processing applications.

The Thai Language Toolkit (TLTK) [5] has developed an algorithm for segmenting the components of Thai words using a rule-based approach to identify the initial consonants, vowels, final consonants, and tone marks sequentially. However, the rule-based system cannot account for all Thai words, particularly those that undergo vowel, or when a consonant serves as one of the vowel sounds. For instance, in the word เตรียม (prepare), the vowel is segmented as เ-, whereas the correct segmentation should be เ-ย.

B. Thai Spelling Distributions

Thai spelling distribution is a fundamental process in phonetic segmentation, systematically analyzing the phonetic components of a word, including its initial consonant, vowel, final consonant, and tonal marker. This process serves as a crucial foundation for early reading development, enabling learners to understand the principles of Thai pronunciation. Thai spelling can be categorized into two types [6]

1. Spelling Distributions for Reading: This method follows the natural phonetic order by sequentially identifying the initial consonant, vowel, final consonant, and tone mark as they appear in pronunciation. For example, the word แพ้ (“lose”) is segmented as พอ แอ แพ ไม้โท แพ้.
2. Spelling Distribution for Writing: This method follows the visual sequence of the word, emphasizing the order in which the characters appear in writing. For instance, แพ้ is segmented as แอ พอ แพ ไม้โท แพ้.

The Spelling Distributions for the Reading method are critical in helping learners grasp pronunciation patterns, as they align closely with phonetic articulation. However, mastering this approach requires understanding Thai tonal rules, including tone variations based on consonant classes, the influence of leading consonants, and the distinction between live and dead syllables. This method is widely used in early childhood education, particularly in first-grade curricula, to build a foundational understanding of phonetic segmentation before transitioning to full-word reading at more advanced levels.

C. Natural Language Processing

Natural Language Processing (NLP) is a crucial field within data science that integrates machine learning models and computational linguistics to enable computers to process, understand, and interpret human language. Given the vast and continuously growing volume of linguistic data in various formats—including text, video, and audio—NLP techniques have evolved significantly to handle complex language structures efficiently. Thai presents unique challenges due to its complex orthography and lack of clear word boundaries.

D. Levensthein Distance

Another widely used technique in NLP is the Levenshtein distance algorithm, also known as the edit distance. This algorithm provides a method for measuring the difference between two text strings—a reference string and a comparison string—by determining the minimum number of operations required to transform the comparison string into the reference string. The possible operations include Insertion, Deletion, Substitution, Transposition [7]

The technique has been applied to cross-lingual word-matching tasks, including Thai. In the study [8], the Levenshtein distance was utilized as a scoring measure to compare pairs of Thai and English names, with both languages being preprocessed into a transliterated format. Additionally, the researchers implemented other similar calculation algorithms, such as the Jaro-Winkler distance. The results indicated that the Levenshtein distance outperformed the Jaro-Winkler distance, achieving 99.98% accuracy.

III. Generation of Thai Spelling Distributions for Reading through the Improvement of Natural Language Processing Techniques

This study presents a method to generate Thai spelling distribution for reading using enhanced NLP with statistical information on NLD. This process aims to decompose Thai word components and use the decomposed components to generate Thai spelling distribution.

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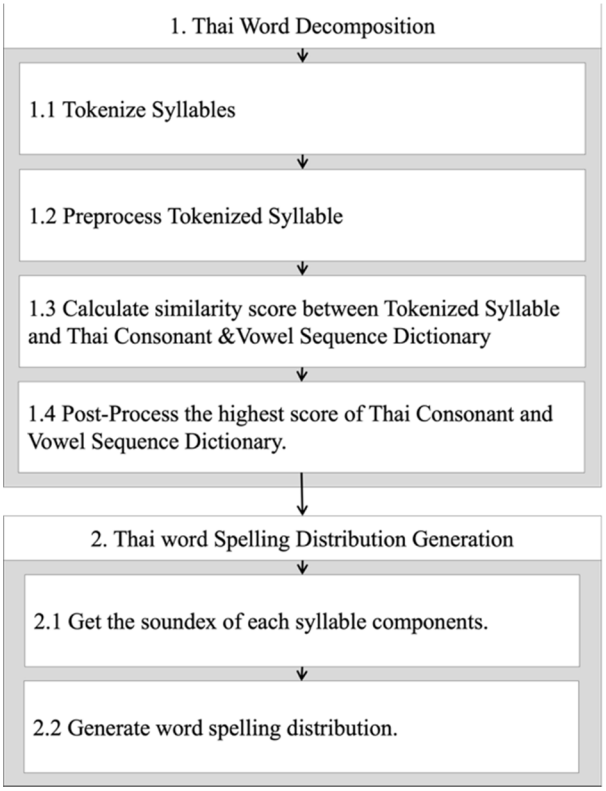


Fig. 1. Methodology for Thai Spelling Distribution Generation.

As shown in Fig. 1, the methodology begins with decomposing Thai words. This step starts with syllabification, where words are segmented into syllables—the minor phonological units in Thai—formed by combining an initial consonant, a vowel, a final consonant, and a tone marker. This step is needed using the Syllable Tokenizer from PyThaiNLP.

This study customizes a dictionary that defines the structural patterns of Thai syllables to classify syllabic components into consonants, vowels, final consonants, and tone markers. The dictionary is used to compute each syllable's most similar sequence pattern using the NLD score. The identified structural patterns are subsequently compared, grouped, and mapped to their corresponding Thai phonemes, ultimately leading to the generation of Thai spelling distributions for reading.

A. Thai Consonant and Vowel Sequence Dictionary

The decomposition of consonant and vowel components within a syllable presents several challenges. Thai syllables may contain one or more initial consonants, while vowels may undergo reductions or transformations or may occur without a final consonant. These factors contribute to the variability in the arrangement of consonants and vowels. This study investigates these structural variations and introduces a method for component extraction using the Thai Consonant and Vowel Sequence Dictionary.

Table I shows that the Thai Consonant and Vowel Sequence Dictionary is divided into two types. Type I is designed for vowel component extraction, where consonant positions—initial and final—are replaced with "-". Type II focuses on consonant classification, distinguishing initial and final consonants by replacing initial consonants with "-" and final consonants with "_." Both types are parallel based on

TABLE I. THAI CONSONANT AND VOWEL SEQUENCE TYPES I AND TYPES II FOR OPEN AND CLOSED VOWELS

Thai Vowel	Thai Consonant and Vowel Sequence			
	Open Vowel		*Close Vowel*	
	Type I	*Type II*	*Type I*	*Type II*
-ะ (a)	-ะ (-a)	-ะ (-a)	ั-- (-a-)	ั_ (-a_)
	- (-a)	- (-a)	-ร- (-a-)	-ร_ (-a_)
-า (aa)	-า (aa)	-า (aa)	-า- (-aa-)	-า_ (-aa_)
แ-ะ (ae)	แ-ะ (-ae)	แ-ะ (-ae)	แ็-- (-ae-)	แ็_ (-ae_)
แ- (aae)	แ- (-aae)	แ- (-aae)	แ-- (-aae-)	แ-_ (-aae_)
เ-าะ (aw)	เ-าะ (-aw)	เ-าะ (-aw)	็อ- (-aw-)	็อ_ (-aw_)
	็ (-aw)	็ (-aw)		
-อ (aaw)	-อ (-aaw)	-อ (-aaw)	-อ- (-aaw-)	-อ_ (-aaw_)
เ-ะ (eh)	เ-ะ (-eh)	เ-ะ (-eh)	เ็-- (-e-)	เ็_ (-e_)
เ- (aeh)	เ- (-aeh)	เ- (-aeh)	เ-- (-aeh-)	เ-_ (-aeh_)
เ-อะ (uh)	เ-อะ (-uh)	เ-อะ (-uh)		
เ-อ (uuhr)	เ-อ (-uuhr)	เ-อ (-uuhr)	เ-- (-eer-)	เ-_ (-eer_)
โ-ะ (o)	โ-ะ (-o)	โ-ะ (-o)	- _ (-_)	- _ (-_)
โ- (o:h)	โ- (-o:h)	โ- (-o:h)	โ-- (-o:h-)	โ-_ (-o:h_)
ิ (i)	ิ (-i)	ิ (-i)	ิ-- (-i-)	ิ_ (-i_)
ี (ee)	ี (-ee)	ี (-ee)	ี-- (-ee-)	ี_ (-ee_)
ึ (eu)	ึ (-eu)	ึ (-eu)	ึ-- (-eu-)	ึ_ (-eu_)
ือ (euu)	ือ (-euu)	ือ (-euu)	ื-- (-euu-)	ื_ (-euu_)
ุ (oo)	ุ (-oo)	ุ (-oo)	ุ-- (-oo-)	ุ_ (-oo_)
ู (uu)	ู (-uu)	ู (-uu)	ู-- (-uu-)	ู_ (-uu_)
เ-ียะ (ia)	เ-ียะ (-ia)	เ-ียะ (-ia)		
เ-ีย (iia)	เ-ีย (-iia)	เ-ีย (-iia)	เ-ีย- (-iia-)	เ-ีย_ (-iia_)
เ-ือะ (eua)	เ-ือะ (-eua)	เ-ือะ (-eua)		
เ-ือ (euua)	เ-ือ (-euua)	เ-ือ (-euua)	เ-ือ- (-euua-)	เ-ือ_ (-euua-)
ั-วะ (ua)	ั-วะ (-ua)	ั-วะ (-ua)		
ั-ว (uaa)	ั-ว (-uaa)	ั-ว (-uaa)	-ว- (-uaa-)	-ว_ (-uaa_)
ำ (am)			ำ (-am)	ำ (-am)
ไ- (ai)			ไ- (-ai)	ไ- (-ai)
ใ- (ai)			ใ- (-ai)	ใ- (-ai)
เ-า (ao)	เ-า (-ao)	เ-า (-ao)		

Thai vowel structures. The dictionary is subsequently applied to syllables processed during preprocessing, and the NLD is computed to identify the most similar pattern. The extracted components are then analyzed and utilized for subsequent syllabic decomposition, as Section C discusses.

B. NLD Calculation

The Levenshtein distance is an algorithm approximating the similarity between word pairs by calculating the minimum edit distance required to transform one string into another. This transformation involves three possible operations: insertions, deletions, and substitutions. The Levenshtein distance is a fundamental tool in Natural Language Processing (NLP) and has been widely applied in various tasks, such as

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Algorithm 1: Syllable Decomposition Algorithm

Result: Thai syllabic components: *initial_consonants, vowels, final_consonants, and tone_marks.*

Input: thai_syllable, sequence_dictionary, *thai_consonant_lists*

1 **Function** DecompositionWord (*thai_syllable*, *sequence_dictionary*) :
2 *tone_mark, syllable* ← Extract and Reomve *tone_mark*
3 *syllable_mask* ← PreprocessSyllable(*syllable, thai_consonant_lists*)
4 compare *syllalble_mask* with sequeces in *sequence_dictionary_type_i*
5 *sequences1* ← find *sequences1* with the highest NLD scores
6 *sequences2* ← get matching *sequences* from *sequence_dictionary_type_ii*
7 **If** multiple *sequences* **do**
8 compare each sequence in sequences2 with syllable using NLD
9 best_sequence_match ← select the sequence with the highest NLD score
10 **end**
11 *vowel, initial_consonants, final_consonants* ← MapExtractVowelConsonants (*syllable, match_sequences2*)
12 **Return** *tone_mark, vowel, initial_consonants, final_consonants*

Algorithm 2: Syllable Preprocessing Algorithm

Result: preprocessed_syllable

Input: syllable, *thai_consonant_lists*

1 **Function** PreprocessSyllable (*syllable, thai_consonant_lists*) :
2 preprocessed_syllable = “”
3 **for** each character in syllables **do**
4 **if** *character* in *thai_consonant_lists* and not in [“ว”, “ย”, “อ”] **do**
5 *preprocessed_syllable* ← *preprocessed_syllable* + *“-”*
6 **else do**
7 *preprocessed_syllable* ← *preprocessed_syllable* + *character*
8 **end**
9 **end**
10 **Return** *preprocessed_syllable*

spelling correction, string matching, and text similarity measurements. [6]

The formula for calculating the Normalized Levenshtein Distance (NLD) is as follows:

$$NLD = \frac{(|a| + |b|) - lev(a,b)}{|a| + |b|} \quad (1)$$

where:

$|a|$ is the length of the first string

$|b|$ is the length of the second string

$lev(a, b)$ is the Levenshtein distance between the two strings

The NLD is a normalized measure of similarity, ensuring that the similarity score for each word pair lies within the same space regardless of word length. This normalization process converts the score into a percentage, making it more suitable for comparing the target word against a list of candidate words from the dictionary, as applied in this research. [8]

C. Thai Word Decomposition

The structure of Thai words can consist of one or more syllables. Segmenting words into syllables provides the minor linguistic units in the Thai language, composed of an initial consonant, a vowel, a final consonant, and a tone marker. In this research, the Thai syllable tokenizer, developed by PyThaiNLP, is employed due to its high efficiency. Each syllable is then decomposed into its components: initial consonants, vowels, final consonants, and tone marks, following the procedure presented in Algorithm 1.

The proposed methodology begins by extracting and removing the tone marker from the syllable. The remaining syllable then undergoes preprocessing as described in Algorithm 2, where all consonants except “ว” (w), “ย” (y), and “อ” (o), which may function as vowel components, are replaced with a "-" to match patterns from the Thai Consonant and Vowel Sequence Type I.

The preprocessed syllable is subsequently compared with each sequence in the Thai Consonant and Vowel Sequence Typeg the syllable (without tone markers and preprocessing) with the candidate sequences from Type II using NLD scores. The sequence with the highest NLD score is selected.

Once the best sequence is identified, the consonant and vowel components are extracted. The process involves mapping "-" and "" to identify initial consonants and final consonants based on the principles of Thai orthography. Special rules apply if a consecutive pair of "-" and "" appears and belongs to one of the consonant clusters. If such a cluster is detected, it is categorized as part of the initial consonant; otherwise, the first consonant is assigned as the initial consonant, and the second consonant is considered the final consonant.

D. Thai Spelling Distribution Generation

The process of generating Thai spelling distributions involves decomposing words into their phonetic components by mapping the sounds of initial consonants, vowels, final consonants, and tone marks according to TABLE II. And TABLE III. The spelling sequence follows this pattern: "Initial consonant sound + vowel sound + final consonant sound (if any) + blended sound of the consonant and vowel + tone mark sound (if present) + blended sound of the consonant, vowel, and tone mark (if present)." For example, the word "ลอง (laawng^{M})" is decomposed into "ลอ - ออ - งอ - ลอง (laaw^{M} - aaw^{M} - ngaaw^{M} - laawng^{M})" or "ก้าง (gaang^{F})" into "กอ - อา - งอ - กาง - ไม้โท - ก้าง (gaaw^{M} - aa^{M} - ngaaw^{M} - gaang^{M} - maai^{H}tho:h^{M} - gaang^{F})." This step-by-step breakdown facilitates a better understanding of Thai phonology, especially for learners or computational models.

If the initial consonant cluster contains more than two consonants and forms a consonant cluster or a leading consonant pair ("ห" leading or "อ" leading "ย"), both consonant sounds are pronounced consecutively before the vowel. For instance, the word "ก ล้ อ ง (glaawng^{F})" is segmented into "กอ - ลอ - ออ - งอ - กลอง - ไม้โท - กล้อง (gaaw^{M} - laaw^{M} - aaw^{M} - ngaaw^{M} - glaawng^{M} - maai^{H}tho:h^{M} - glaawng^{F})." This rule highlights the influence of consonant clusters on Thai pronunciation.

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TABLE II. CONSONANT AND VOWEL SOUND MAPPING

Consonant	Soundex (Thai / Romanized)	Vowel	Soundex (Thai / Romanized)
ก	กอ gaaw^{M}	-ะ	อะ a^{L}
ข ฃ	ขอ khaaw^{R}	-า	อา aa^{M}
ค ฅ ฆ	คอ khaaw^{M}	แ-ะ	แอะ ae^{L}
ง	งอ ngaaw^{M}	แ-	แอ aae^{M}
จ	จอ jaaw^{M}	เ-าะ	เอาะ aw^{L}
ฉ	ฉอ chaaw^{R}	-อ	ออ aaw^{M}
ช ฌ	ชอ chaaw^{M}	เ-ะ	เอะ eh^{L}
ซ	ซอ saaw^{M}	เ-	เอ aeh^{M}
ญ ย	ยอ yaaw^{M}	เ-อะ	เออะ uh^{L}
ฎ ด	ดอ daaw^{M}	เ-อ	เออ uuhr^{M}
ฏ ต	ตอ dtaaw^{M}	โ-ะ	โอะ o^{L}
ฐ ถ	ถอ thaaw^{R}	โ-	โอ o:h^{M}
ฑ ฒ ท ธ	ทอ thaaw^{M}	ิ-	อิ i^{L}
ณ น	นอ naaw^{M}	ี-	อี ee^{M}
บ	บอ baaw^{M}	ึ-	อึ eu^{L}
ป	ปอ bpaaw^{M}	ื-อ	อือ euu^{M}
ผ	ผอ phaaw^{R}	ุ-	อุ oo^{L}
ฝ	ฝอ faaw^{R}	ู-	อู uu^{M}
พ ภ	พอ phaaw^{M}	เ-ียะ	เอียะ ia^{L}
ฟ	ฟอ faaw^{M}	เ-ีย	เอีย iia^{M}
ม	มอ maaw^{M}	เ-ือะ	เอือะ eua^{L}
ร	รอ raaw^{M}	เ-ือ	เอือ euua^{M}
ล ฬ	ลอ laaw^{M}	ั-วะ	อัวะ ua^{L}
ว	วอ waaw^{M}	ั-ว	อัว uaa^{M}
ศ ษ ส	สอ saaw^{R}	ำ	อำ am^{L}
ห	หอ haaw^{R}	ไ-	ไอ ai^{M}
อ	ออ aaaw^{M}	ใ-	ใอ ai^{M}
ฮ	ฮอ haaw^{M}	เ-า	เอา ao^{M}

TABLE III. TONE MARK MAPPING IN ROMANIZED THAI

Tone Mark	Romanized Soundex (Thai / Romanized)	Tone Mark	Romanized Soundex (Thai / Romanized)
่	ไม้เอก maai^{H} aehk^{L}	๎	ไม้ตรี maai^{H} dtree^{M}
้	ไม้โท maai^{H} tho:h^{M}	๋	ไม้จัตวา maai^{H} jat^{L} dta^{L} waa^{M}

If the consonant cluster does not belong to these categories, the word is treated as a two-syllable word with a reduced short vowel "-ะ" and decomposed as follows: "สตูล (sa^{L}dtuun^{M})" becomes "สอ - อะ - สะ - ตอ - อู - ลอ - ตูล - สตูล (saaw^{R} - a^{L} - sa^{L} - dtaaw^{M} - uu^{M} - laaw^{M} - dtuun^{M} - sa^{L}dtuun^{M})."

For consonant-vowel and consonant-vowel-tone mark combinations in Romanized Thai, it is necessary to consider whether the syllable contains a leading consonant, is alive or dead syllable, and whether the initial consonant is high, middle, or low class. The tone is then mapped to the appropriate Romanized tone markers: Low (L), Mid (M), High (H), Falling (F), or Rising (R), as shown in TABLE IV.

TABLE IV. TONE SOUND MAPPING RULES

Tone Mark	Leading Consonant	High-Class Initial	Mid-Class Initial	Low-Class Initial
-	No	R (live), L (dead)	M (live), L (dead)	M (live), F (dead)
	Yes	-	-	R (live), L (dead)
่	No	L	L	F
	Yes	-	-	L
้	No	F	F	H
	Yes	-	-	F
๊	Any	-	H	-
๋	Any	-	R	-

The test dataset can be categorized according to the following Thai word characteristics:

1) Classification by Initial Consonants

a) Single Initial Consonant (SIC): Words with a single initial consonant, totaling 647 words.

b) Consonant Clusters (CC): Words with two initial consonants forming legitimate consonant clusters, This dataset includes only true consonant clusters, with 33 words.

c) Consonant Leading Pairs (CL): Word with two initial consonants forming leading consonant pairs, including "**ห นำ** (h-nam)" with visible and invisible "**ห**" as well as "**อ นำ ย** (o-nam-ya)" There are 28 words in this category.

2) Classification by Vowels

a) Regular Vowels (regV): Words containg standard vowel forms totaling 559 words.

b) Reduced Vowels (redV): Words with reduced vowel forms, totaling 59 words.

c) Transformed Vowels (transV): Words with transformed vowel forms, totaling 90 words.

3) Classification by Final Consonants

a) No final Consonant (nFC): Words without a final consonant, totaling 200 words.

b) Regular Final Consonants (rFC): Words with regular final consonants that adhere to the standard Thai consonant pattern, totalling 456 words.

c) Irregular Final Consonants (irFC): Word with irregular final consonants that deviate from the standard Thai consonant pattern, totalling 2 words.

As shown in Fig. 2, the proposed method for segmenting word components from the test set in each category, when compared to Human Annotation, demonstrates almost equally accurate decomposition of word components across all types of words.

E. Thai Spelling Distribution Generation

The proposed method for generating Thai spelling distributions, based on the identified word and syllable components, demonstrates high accuracy.

The PyThaiNLP library provides a word_syllable function that focuses on spelling syllables using simple NLP methods to classify phonetics and map to spelling distributions. Example results of each method are shown in TABLE V.

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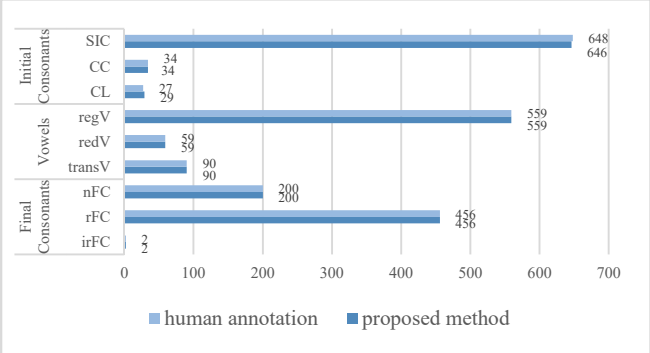


Fig. 2. Comparison in Word Decomposition between Human Annotation and Proposed Method

TABLE V. EXAMPLES OF THAI AND ROMANIZED SPELLING DISTRIBUTIONS

Thai Word	PyThaiNLP	Ours Approach	
	Spelling Distribution in Thai	Spelling Distribution in Thai	Spelling Distribution in Romanized Form
ฟีน	ฟอ-อีน-ฟีน	ฟอ-อือ-นอ-ฟีน	faaw^{M}-euu^{M}-naaw^{M}-feuun^{M}
สับปะรด	สอ-อะ-บอ-สับ-ปอ-อะ-ปะ-รอ-ดอ-รด-สับ-ปะ-รด	สอ-อะ-บอ-สับ-ปอ-อะ-ปะ-รอ-โอะ-ดอ-รด-สับ-ปะ-รด	saaw^{R}-a^{L}-baaw^{M}-sap^{L}-bpaaw^{M}-a^{L}-bpa^{L}-raaw^{M}-o^{L}-daaw^{M}-rohd^{H}-sap^{L}-bpa^{L}-rohd^{H}
ตะไคร้	ตอ-อะ-ตะ-คอ-ไอ-รอ-ไคร-ไม้โท-ไคร้-ตะ-ไคร้	ตอ-อะ-ตะ-คอ-รอ-ไอ-ไคร-ไม้โท-ไคร้-ตะ-ไคร้	dtaaw^{M}-a^{L}-dta^{L}-khaaw^{M}-raaw^{M}-ai^{M}-khrai^{M}-mai^{H}-tho:h^{M}-khrai^{H}- dta^{L}- khrai^{H}
หนา	หอ-นอ-อา-หนา	หอ-นอ-อา-หนา	haaw^{R}-naaw^{M}-aa^{M}-naa^{R}
ขยะ	ขอ-ยอ-อะ-ขยะ	ขอ-ยอ-อะ-ขยะ	khaaw^{R}-yaaw^{M}-a^{L}-kha^{L}-ya^{H}
อยู่	ออ-ยอ-อู-ยู-ไม้เอก-อยู่	ออ-ยอ-อู-อยู-ไม้เอก-อยู่	aaw^{M}-yaaw^{M}-uu^{M}-yuu^{R}-mai^{H}-aehk^{L}-yuu^{F}

We compare our approach with PyThaiNLP's word_syllable function using a test dataset. In terms of accuracy, our method significantly outperforms PyThaiNLP, achieving 99.7% accuracy compared to PyThaiNLP's 87.0%. The accuracy gap is especially evident in challenging categories such as redV and transV, where PyThaiNLP often misclassifies phonetic components like initial consonants, final consonants, and vowels.

In terms of processing speed, our method also demonstrates superior performance, with an average processing time of 59.7 µs—making it approximately 2.38 times faster than PyThaiNLP, which averages 142 µs.

IV. CONCLUSION AND FUTURE WORK

This study presents a method for segmenting Thai words into syllabic components—initial consonants, vowels, tone marks, and final consonants—using a Thai Consonant and Vowel Sequence Dictionary, which stores vowel pairs and consonant-vowel patterns based on linguistic rules. The method calculates the closest consonant-vowel sequence to a target syllable using the Normalized Levenshtein Distance (NLD) score, facilitating efficient component grouping that simulates human annotation.

The extracted components are also mapped to phonetic sounds to generate Thai and Romanized spelling distributions, thereby simplifying the learning of spelling patterns for early readers. Our approach not only demonstrates superior accuracy in generating Thai spelling distributions compared to existing methods such as PyThaiNLP, but also achieves faster processing speeds, making it more suitable for real-time or large-scale applications.

While the method effectively supports introductory-level words, its applicability to advanced vocabulary, such as loanwords or royal terms, may be limited. Future work will focus on developing these distributions as ground-truth data for integration with a Romanized speech-to-text model, aiming to create an application to assist teachers in evaluating students' reading and pronunciation skills.

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